

JC4004 – Computational Intelligence

Sessions 9 & 10: Fuzzy logic

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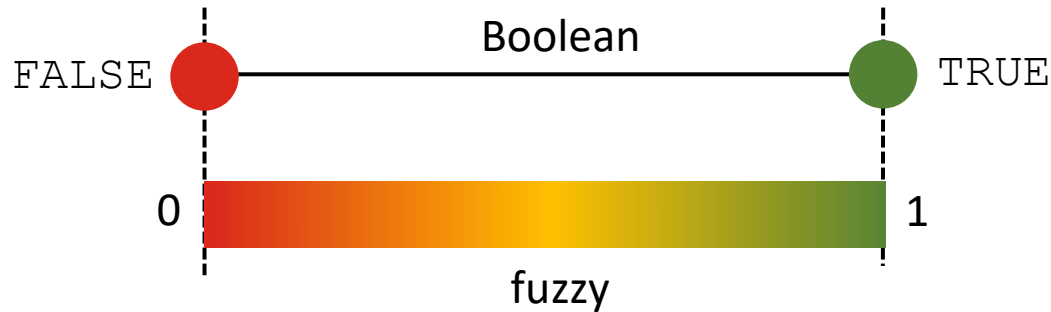
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What is fuzzy logic?

- Fuzzy logic is an alternative to traditional binary Boolean logic to deal with reasoning that is approximate rather than fixed and exact
 - Many concepts in fuzzy logic follow Boolean logic, but taking the uncertainties and inaccuracies of real-life phenomena into account
- Fuzzy logic allows to control system using vague linguistic rules such as *“if the temperature is hot, increase the fan speed”*
 - Numerical values are mapped to fuzzy linguistic definitions and vice versa
- The next two sessions will introduce the fundamentals of fuzzy logic

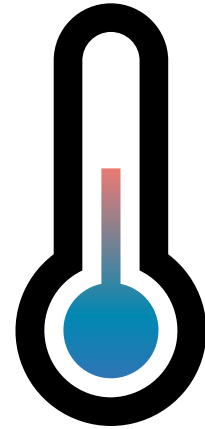
Fuzzy logical variables

- In traditional Boolean logic, logical variables have two possible values: FALSE (0) and TRUE (1)
- However, many real-life concepts and phenomena cannot be defined precisely: to express uncertainties and different degrees of truth, fuzzy logic uses logical variables with values ranging from 0 to 1



Logical variable example

- Let us assume that we need to describe the outside temperature using logical variables *cold* and *hot*, and the temperature is +20°C
- Most people would agree that +20°C is neither cold nor hot, but something between
 - However, in Boolean logic, we need to choose either *cold*=TRUE and *hot*=FALSE, or *cold*=FALSE and *hot*=TRUE
 - In fuzzy logic, we could assign *cold*=0.5 and *hot*=0.5



Fuzzy logical operators

- *Fuzzy logical operators* try to mimic Boolean logical operators as long as possible; common replacements are *Zadeh* and *probabilistic* operators:

Boolean	Zadeh fuzzy	probabilistic fuzzy
$x \text{ AND } y$	$\text{MIN}(x,y)$	$x \cdot y$
$x \text{ OR } y$	$\text{MAX}(x,y)$	$x + y - x \cdot y$
$\text{NOT } x$	$1 - x$	$1 - x$

- Note that when $x=0$ or $x=1$, and $y=0$ or $y=1$, Boolean and fuzzy operators give the same result!

Fuzzy logical operation example

- Statement: ***If the weather is warm and not rainy, I will go for a walk.***
- Let us assume that it is raining slightly (drizzling), and the temperature is +25°C; in Boolean terms, *rainy* = TRUE and *warm* = TRUE

warm AND (NOT *rainy*) = TRUE AND (NOT TRUE) = FALSE

- Result: ***I will not go for a walk.***
- In fuzzy terms, we could assign *rainy* = 0.4 and *warm* = 0.9

Zadeh: $\text{MIN}(\text{warm}, 1 - \text{rainy}) = \text{MIN}(0.9, 1 - 0.4) = \underline{0.6}$

Probabilistic: $\text{warm} \cdot (1 - \text{rainy}) = 0.9 \cdot (1 - 0.4) = \underline{0.54}$

- Result: ***I will go for a short walk.***

Fuzzy set

- *Fuzzy set* is a pair (U, m) , where U is a set and m is a membership function with range $[0,1]$, defining degrees of membership
- Given that $x \in U$, we can say that:

x is not included (no member) in (U, m) if $m(x) = 0$

x is fully included (full member) if $m(x) = 1$

x is partially included (fuzzy member) if $0 < m(x) < 1$

- Note that a classical “crisp” set is a special case of a fuzzy set, where $m(x) = 1 \forall x \in U$

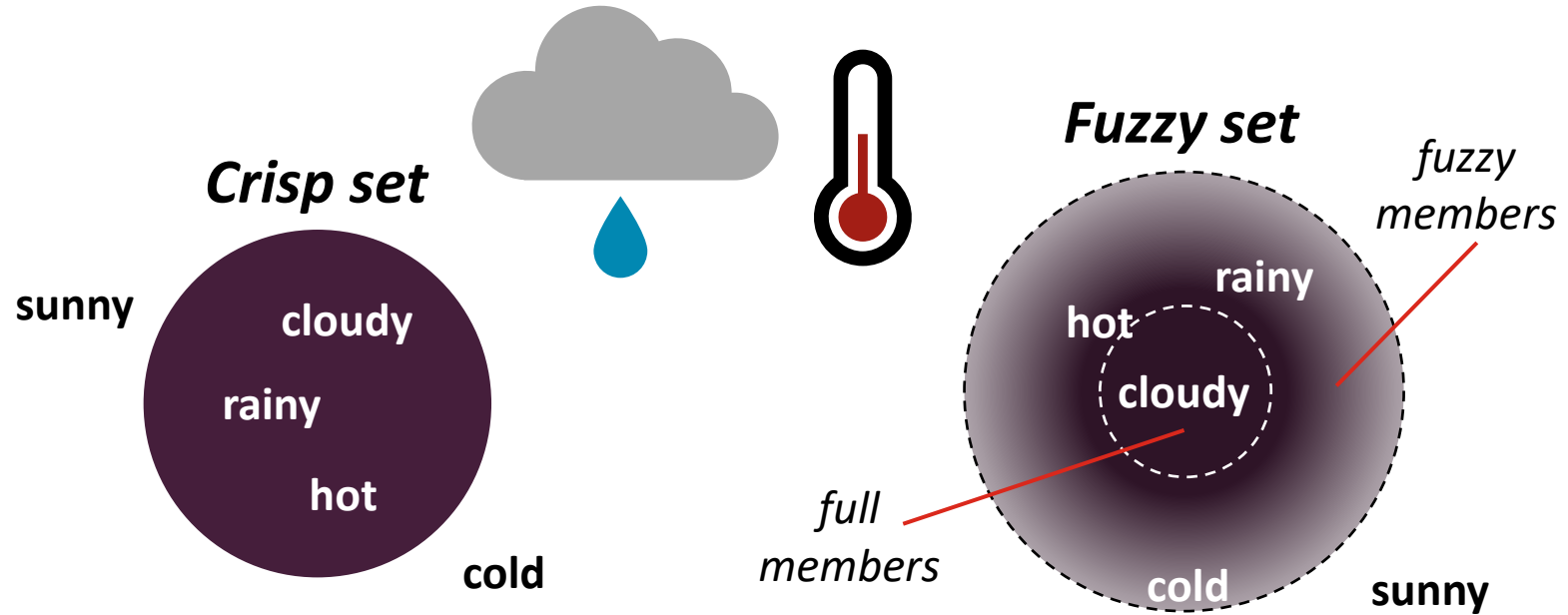
Fuzzy set example

- Let us assume that we want to define the current weather conditions using a fuzzy set with members $U = \{ \text{rainy}, \text{cloudy}, \text{sunny}, \text{hot}, \text{cold} \}$
- Given that it is raining slightly, the sky is fully cloudy, and the temperature is $+25^{\circ}\text{C}$ we could define the membership function as:

$m(\text{rainy})$	$= 0.5$
$m(\text{cloudy})$	$= 1$
$m(\text{sunny})$	$= 0$
$m(\text{hot})$	$= 0.9$
$m(\text{cold})$	$= 0.1$

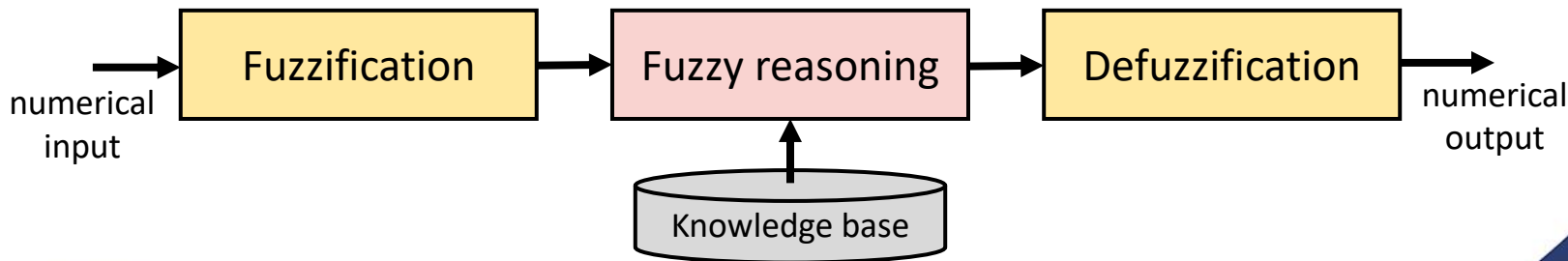


Crisp vs. fuzzy set



Mamdani fuzzy system

- Mamdani fuzzy systems contain three parts:
 - *Fuzzification*: assign values to the membership functions for fuzzy reasoning, according to the system input
 - *Fuzzy reasoning*: apply the rules from the fuzzy *knowledge base* to the fuzzy input set to perform the desired task of the system
 - *Defuzzification*: convert the fuzzy results to crisp output values



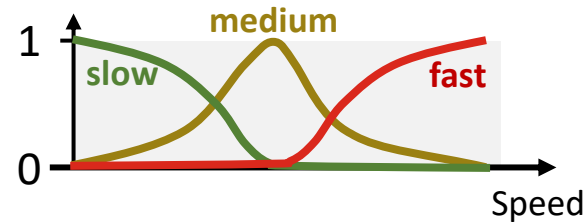
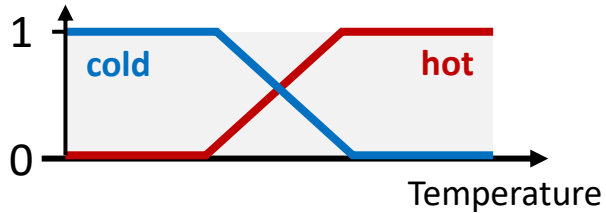
Linguistic variables

- In fuzzy systems, *linguistic variables* are fuzzy sets used to apply fuzzy rules in natural language
 - Each linguistic variable has a descriptive name (e.g., “temperature” or “speed”) and a fuzzy set with members representing the qualities of the variable in natural language, such as “cold”, “hot”, “slow”, and “fast”
 - Each linguistic variable also defines a range of possible input values (*universe of discourse*) to be mapped to the fuzzy set

Example: name: *outside_temperature*
 universe of discourse: $[-70^{\circ}\text{C}, +60^{\circ}\text{C}]$
 fuzzy set: { *cold, warm, hot* }

Fuzzification

- Fuzzification takes crisp numerical input values and map them into degrees of truth in one or more linguistic variables
 - Input values typically represent real-world measurements, such as temperature, speed, or pressure
 - Mapping is often defined as triangles or trapezoid-shaped curves, but other shapes could be also be used



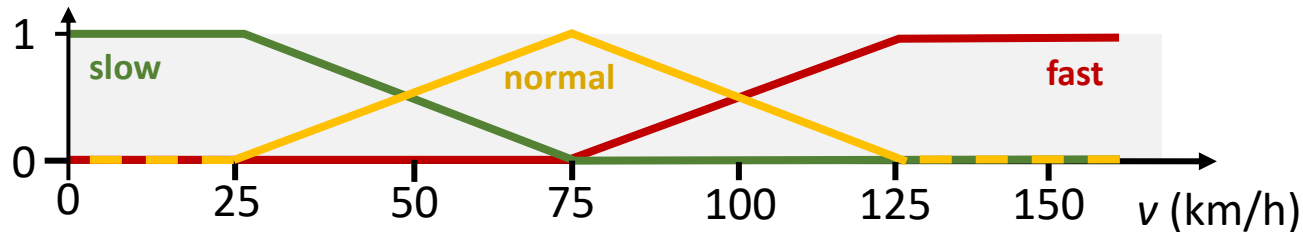
Fuzzification example

- Let us define car speed using fuzzy set { *slow*, *normal*, *fast* }
 - Based on common sense, we can fuzzify the input using the measured velocity v (in km/h) as follows:

$$\begin{aligned} m(\text{slow}) &= 1, & \text{if } 0 \leq v < 25 \\ m(\text{slow}) &= \frac{75-v}{50}, & \text{if } 25 \leq v < 75 \\ m(\text{slow}) &= 0, & \text{if } v \geq 75 \end{aligned}$$

$$\begin{aligned} m(\text{fast}) &= 0, & \text{if } v < 75 \\ m(\text{fast}) &= \frac{v-75}{50}, & \text{if } 75 \leq v < 125 \\ m(\text{fast}) &= 0, & \text{if } v \geq 125 \end{aligned}$$

$$m(\text{normal}) = 1 - m(\text{slow}) - m(\text{fast})$$



Fuzzy inference

- In fuzzy reasoning, fuzzy rules are applied to generate fuzzy output from fuzzy input
- Fuzzy inference rules are written in form of if-then statements

IF x is a THEN y is b

antecedent **consequent**

- x is a linguistic variable used as input and a is its fuzzy subset
- y is a linguistic variable used as output and b is its fuzzy subset
- The truth value of the antecedent is assigned to the consequent

Fuzzy reasoning with logical operators

- Logical operators AND, OR, and NOT can also be used in the antecedent

IF x is a AND y is b THEN z is c

antecedent consequent

- The fuzzy input value can be resolved by replacing the logical operators with their fuzzy replacements (e.g., Zadeh operators)
- As there are different alternative operators, system designer must choose the used operators

Fuzzy reasoning examples

- **Example 1:** IF *temperature* is *hot* THEN *fan_speed* is *high*
 - If it is true that temperature is hot (i.e., the degree of truth of *hot* is large), the fan speed will also be high

$$m_{fan_speed}(high) = m_{temperature}(hot)$$

- **Example 2:** IF *speed* is *fast* AND *road* is *wet* THEN *risk* is *high*
 - We can choose e.g., Zadeh operator to compute AND equivalent, and then assign the result to the consequent

$$m_{risk}(high) = \min(m_{speed}(fast), m_{road}(wet))$$

Fuzzy reasoning with several rules

- Fuzzy systems can implement several fuzzy rules
- Different rules can involve the same or different linguistic variables

Rule 1: IF x is a AND y is b THEN z is c

Rule 2: IF q is d THEN r is e

Rule 3: IF q is f AND x is g THEN z is h

- The consequent of a fuzzy rule can include multiple parts

IF x is a THEN y is b ; z is c

two consequents

Break: any questions?

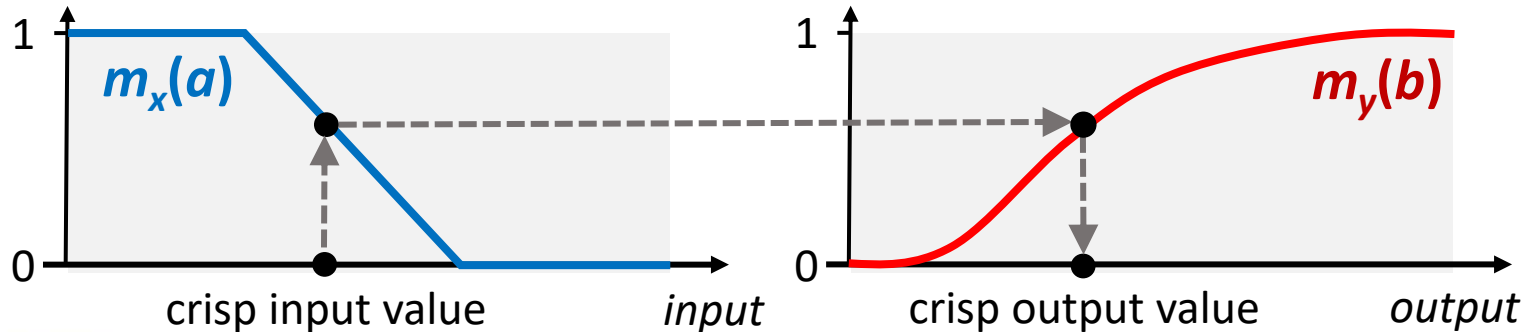
Defuzzification

- Defuzzification is the process of obtaining the crisp output values from the fuzzy linguistic output variables
- There is no single commonly accepted method for defuzzification, several different methods have been proposed
- We will introduce the most commonly used methods:
 - Mean of Maxima (MoM)
 - Center of Gravity (CoG)
 - Center of Area (CoA)

Monotonic selection

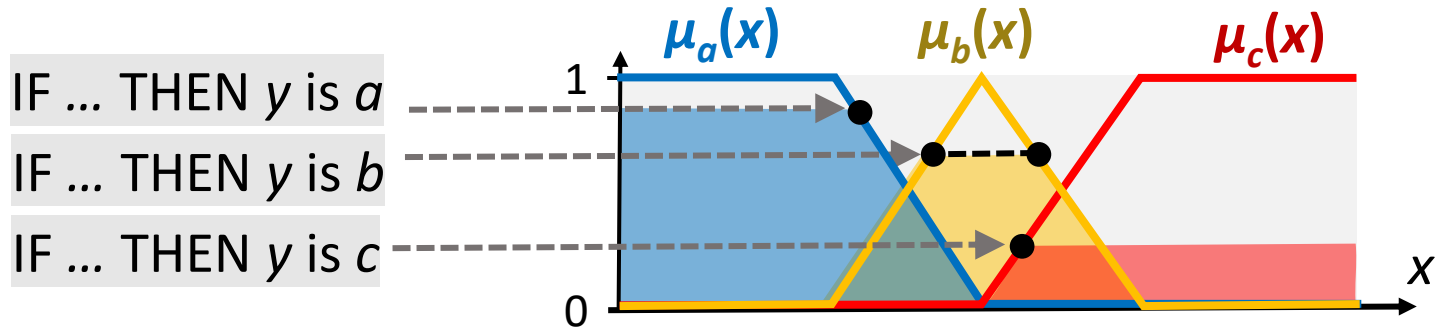
- In a trivial case, there is only one fuzzy output respective to the fuzzy input that can be converted using monotonic mapping
 - In this case, we can just choose the crisp output value that corresponds to the degree of truth of the fuzzy output

IF x is a THEN y is b



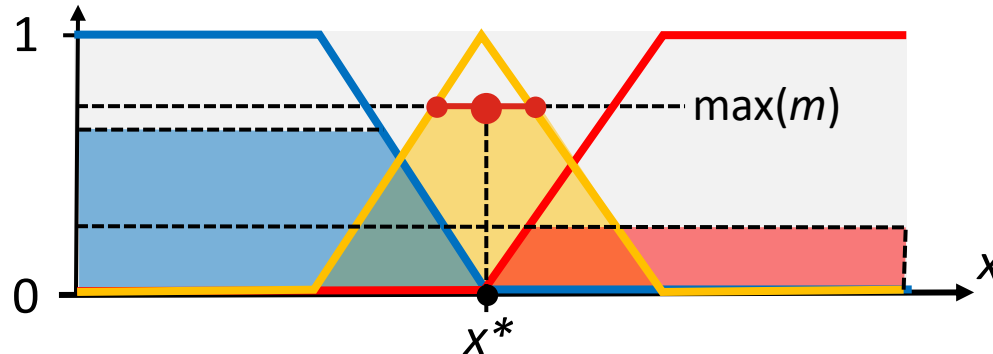
Defuzzifying multiple fuzzy members

- In a typical case, there are multiple fuzzy rules resulting several different fuzzy outputs involving the same fuzzy sets
 - In this case, several degrees of truth are assigned to the same output, so they cannot be converted to crisp output directly



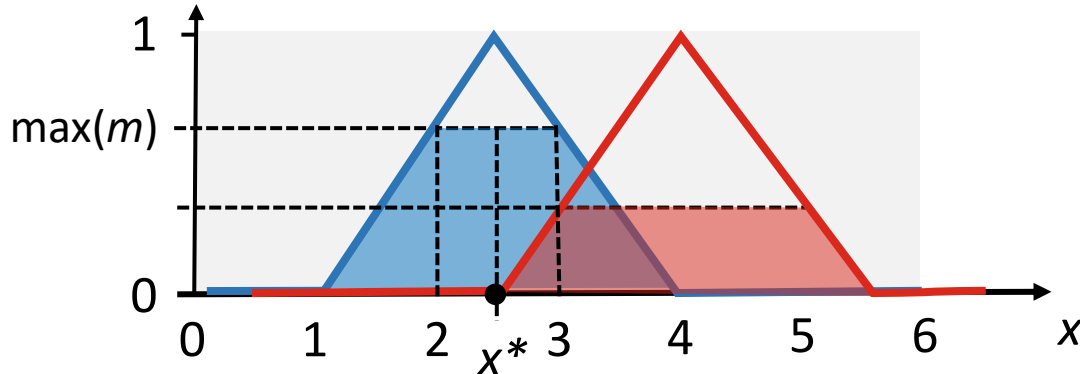
Mean of maxima (MoM) method

- In MoM method, first find the points (or range) where the degree of membership reaches its maximum
- Then, compute the mean of the output values x corresponding to the points with maximum degree of membership to obtain x^*



MoM example

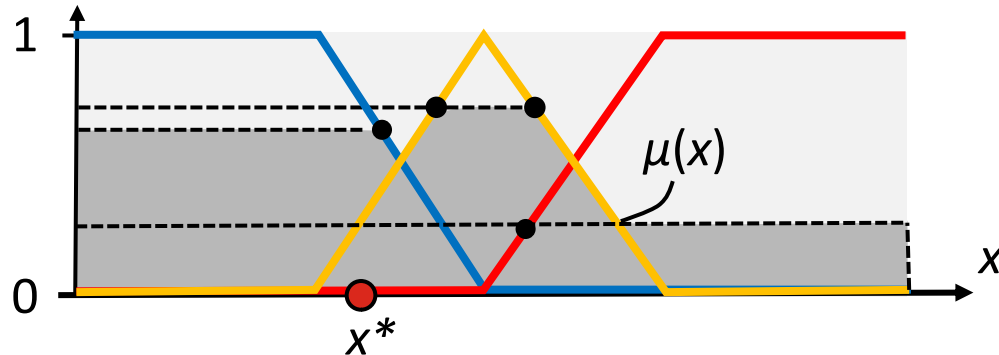
- Given the example below, the highest degree of membership is obtained in the range $x \in [2.0, 3.0]$
- Since the maxima form a continuous range, the mean of the maxima is in the middle, i.e., $x^* = (2.0 + 3.0) = \underline{\underline{2.5}}$



Center of gravity (CoG) method

- In CoG (centroid) method, the fired areas below all the membership functions are considered jointly to compute the center of gravity

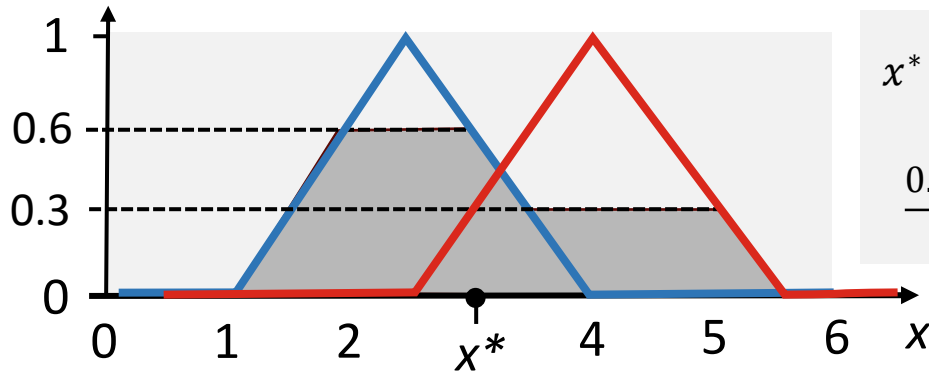
$$x^* = \frac{\int_x x \cdot \mu(x) dx}{\int_x \mu(x) dx}$$



CoG example

- Given the example below, the degree of membership is defined piecewise as:

$$\mu(x) = \begin{cases} 0.6x - 0.6 & \text{if } 1 \leq x < 2; & 0.6 & \text{if } 2 \leq x < 3 \\ 2.4 - 0.6x & \text{if } 3 \leq x < 3.5; & 0.3 & \text{if } 3.5 \leq x < 5 \\ 3.3 - 0.6x & \text{if } 5 \leq x \leq 5.5; & 0 & \text{if } x < 1 \text{ or } x > 5.5 \end{cases}$$

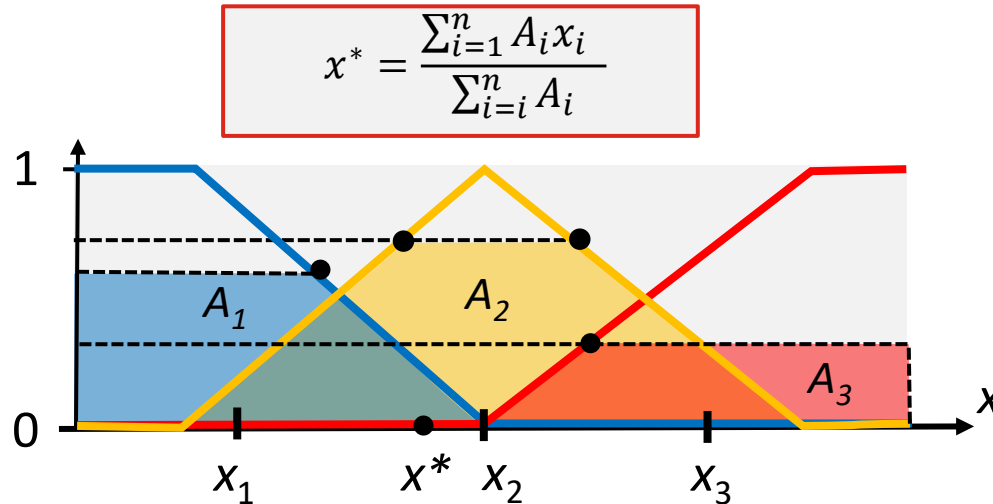


$$x^* = \frac{\int_x x\mu(x)dx}{\int_x \mu(x)dx} = \dots =$$

$$\frac{0.5 + 1.5 + 0.725 + 1.9125 + 0.3875}{0.3 + 0.6 + 0.225 + 0.45 + 0.075} \approx \underline{\underline{3.05}}$$

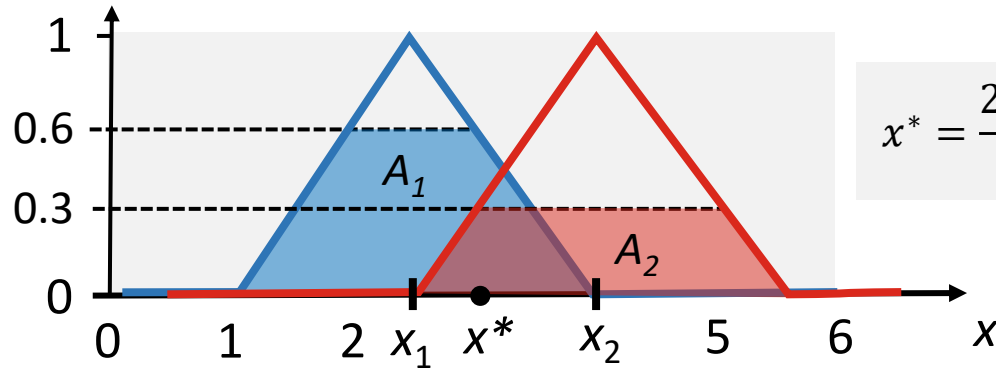
Center of area (CoA) method

- CoA method is similar with CoG method, except that the centroids are computed separately for each area and then combined
 - Easier to compute than CoG, but includes overlapping areas twice



CoA example

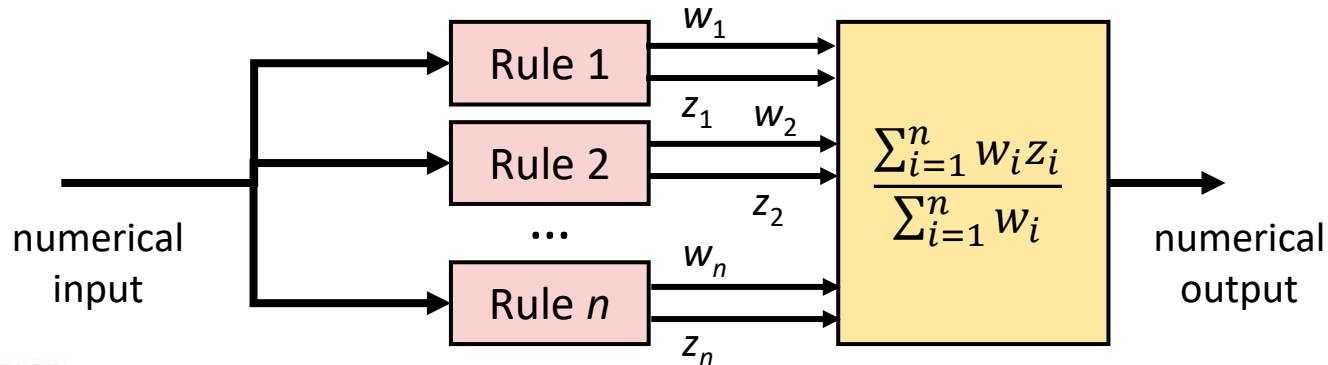
- $A_1 = \frac{1}{2}((4 - 1) + (3 - 2)) \cdot 0.6 = 1.2$
- $A_2 = \frac{1}{2}((5.5 - 2.5) + (5 - 3)) \cdot 0.3 = 0.75$
- The shapes are symmetric, so: $x_1 = \frac{1+4}{2} = 2.5$ and $x_2 = \frac{2.5+5.5}{2} = 4$



$$x^* = \frac{2.5 \cdot 1.2 + 4 \cdot 0.75}{1.2 + 0.75} \approx \underline{\underline{3.08}}$$

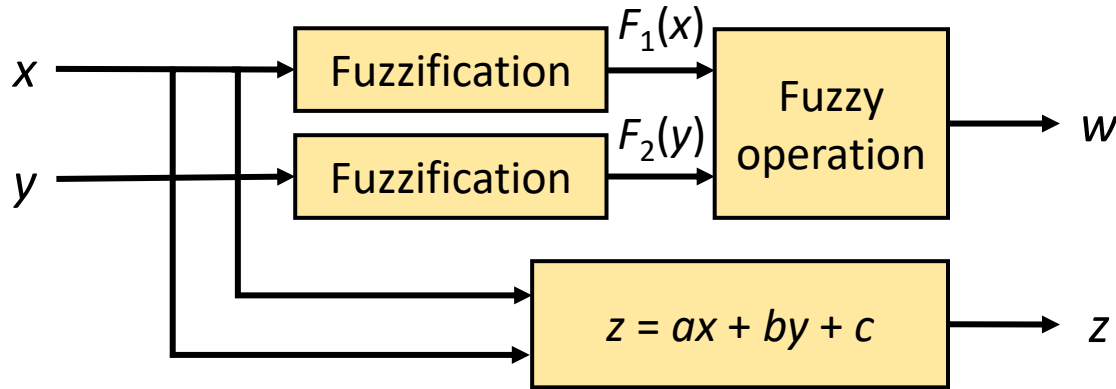
Sugeno fuzzy system

- *Sugeno fuzzy inference system*, also known as Takagi-Sukeno-Kang (TSK) system, combines fuzzy rules and defuzzification
 - Computationally more efficient than Mamdani systems, but the fuzzy rules are less intuitive and interpretable
 - Final output computed as weighted sum of the outputs generated by rules



Fuzzy rules Sugeno system

- Fuzzy rules in Sugeno system generate both output and weight
 - Output z is either a constant or a linear function of the inputs
 - Weight w is computed by applying the specified fuzzy operation(s) to the fuzzified input values



Sugeno system example

- Let us assume input variables *traffic* and *road*, and output variable *speed*, and the following rules:
 - Rule 1: IF *traffic* is *calm* AND *road* is *dry* THEN *speed* = 100
 - Rule 2: IF *traffic* is *busy* THEN *speed* = 70
 - Rule 3: IF *road* is *wet* THEN *speed* = 50
- Let us assume the memberships are assigned as:
Traffic: $m(\text{calm})=0.2$, $m(\text{busy}) = 0.8$, *Road*: $m(\text{dry})=0.6$, $m(\text{wet})=0.4$
- Apply the rules and combine the outputs to get the final output:
$$\text{speed} = (\text{MIN}(0.2,0.6) \cdot 100 + 0.8 \cdot 70 + 0.4 \cdot 50) / (0.2+0.8+0.4) \approx \underline{\underline{68.6}}$$

Sugeno system example

- Let us assume input variables *traffic* and *road*, and output variable *speed*, and the following rules:

Rule 1: IF *traffic* is *calm* AND *road* is *dry* THEN *speed* = 100

Rule 2: IF *traffic* is *busy* THEN *speed* = 70

Rule 3: IF *road* is *wet* THEN *speed* = 50

- Let us assume the following membership values are assigned as:

Traffic: $m(\text{calm})=0.2$, $m(\text{busy})=0.8$, *Road*: $m(\text{dry})=0.6$, $m(\text{wet})=0.4$

- Apply the rules and combine the outputs to get the final output:

$$\text{speed} = (\text{MIN}(0.2, 0.6) \cdot 100 + 0.8 \cdot 70 + 0.4 \cdot 50) / (0.2 + 0.8 + 0.4) \approx \underline{\underline{68.6}}$$

Weights w

Constants z

Summary

- Fuzzy logic allows applying logical rules on vaguely defined linguistic attributes in a similar fashion as in traditional Boolean logic
- Mamdani fuzzy systems consist of different parts for fuzzification, fuzzy rules, and defuzzification
 - Fuzzification is the process of mapping crisp input to degrees of truth
 - Fuzzy rules used to infer logical operations on the fuzzy input
 - Defuzzification is the process of mapping fuzzy output to crisp output values: different defuzzification methods are available
- Sugeno fuzzy systems combine logical rules and defuzzification
 - More efficient than Mamdani systems, but the rules are less intuitive

Questions and discussion